

Kalman Filter Based Data Analysis for UAV Navigation Systems

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Abstract

This study presents a systematic implementation and comparative evaluation of a discrete-time scalar Linear Kalman Filter applied to multi-channel sensor data collected from an Unmanned Aerial Vehicle (UAV) navigation dataset. Raw sensor readings acquired from Inertial Measurement Units (IMUs), GPS modules, LiDAR sensors, and auxiliary peripherals are inherently corrupted by measurement noise, sensor drift, and environmental disturbances, which collectively degrade the reliability and accuracy of real-time state estimation. The proposed filtering framework independently processes thirteen sensor channels — including latitude, longitude, altitude, tri-axial accelerometer, tri-axial gyroscope, LiDAR distance, vehicle speed, wind speed, and battery level — through a recursive predict-correct cycle executed at each discrete time step k .

The filter operates through two sequential phases at every time step. In the *prediction phase*, the prior state estimate x_k^- and prior error variance P_k^- are propagated forward using:

$$x_k^- = \hat{x}_{k-1} \quad (1)$$

$$P_k^- = P_{k-1} + Q \quad (2)$$

where Q denotes the process noise variance, which governs how rapidly the filter's uncertainty grows between consecutive measurements. In the *correction phase*, the Kalman gain K_k is computed as the optimal weighting coefficient that balances the relative trust between the prediction and the incoming sensor measurement:

$$K_k = \frac{P_k^-}{P_k^- + R} \quad (3)$$

where R is the measurement noise variance characterising the sensor's hardware uncertainty. The posterior state estimate $\hat{x}_k$ — the final filtered output — is then obtained by correcting the prior estimate with a scaled fraction of the innovation $\nu_k = z_k - x_k^-$:

$$\hat{x}_k = x_k^- + K_k \cdot (z_k - x_k^-) \quad (4)$$

$$P_k = (1 - K_k) \cdot P_k^- \quad (5)$$

The posterior error variance P_k strictly decreases after each measurement update, reflecting the monotonic improvement in estimation confidence as more sensor readings are incorporated.

Category-specific noise profiles are configured for each sensor type to reflect their distinct physical noise characteristics. GPS channels are assigned $Q = 10^{-5}$ and $R = 5 \times 10^{-2}$, yielding a low steady-state Kalman gain that produces highly smooth position estimates resistant to multipath-induced jumps. Accelerometer channels are assigned $Q = 10^{-3}$ and $R = 5 \times 10^{-1}$, allowing the filter to track rapid inertial dynamics more responsively. This category-aware parameterisation ensures that the ratio Q/R — the decisive tuning parameter — is appropriately calibrated to the noise regime of each sensor modality.

Filter performance is quantified using two complementary evaluation metrics. The absolute Noise Reduction:

$$\text{Noise Reduction} = \sigma_{\text{raw}} - \sigma_{\text{filtered}} \quad (6)$$

measures the reduction in signal standard deviation achieved by the filter, providing a direct indicator of noise attenuation. The SNR Improvement:

$$\text{SNR Improvement (dB)} = 20 \log_{10} \left(\frac{\text{RMSE}_{\text{raw}}}{\text{RMSE}_{\text{filtered}} + \epsilon} \right) \quad (7)$$

provides a logarithmic, scale-independent measure of signal quality enhancement across sensor channels whose physical units and magnitude ranges differ significantly. An interactive multi-panel visualisation platform is developed alongside the filter to facilitate real-time side-by-side comparison of raw and filtered signals, residual (innovation) analysis, and Kalman gain evolution over time.

Experimental results confirm that the scalar Kalman filter effectively attenuates high-frequency measurement noise across all thirteen sensor channels while faithfully preserving the underlying navigational trajectory dynamics. The findings demonstrate that proper configuration of the Q/R ratio is critical to achieving optimal filter performance, and that the developed visualisation framework provides a powerful diagnostic tool for validating filter behaviour in UAV state estimation applications.

Keywords: Kalman Filter, UAV Navigation, State Estimation, Sensor Fusion, Noise Reduction, IMU, GPS, LiDAR, Signal Processing

Notations

The following symbols and abbreviations are used consistently throughout this study.

Symbol	Definition
<i>Indices & Time</i>	
k	Discrete time step index
$k - 1$	Previous discrete time step
<i>State Variables</i>	
x_k^-	Prior (predicted) state estimate at time step k
$\hat{x}_k$	Posterior (corrected) state estimate at time step k
$\hat{x}_{k-1}$	Posterior state estimate at previous time step $k - 1$
z_k	Raw sensor measurement (observation) at time step k
<i>Error Variance</i>	
P_k^-	Prior error variance at time step k
P_k	Posterior error variance at time step k
P_{k-1}	Posterior error variance at previous time step $k - 1$
<i>Noise Parameters</i>	
Q	Process noise variance (system model uncertainty)
R	Measurement noise variance (sensor hardware uncertainty)
Q/R	Noise ratio; the primary tuning parameter of the filter
<i>Kalman Filter Quantities</i>	
K_k	Kalman gain at time step k
ν_k	Innovation (residual); $\nu_k = z_k - x_k^-$
<i>Performance Metrics</i>	
σ_{raw}	Standard deviation of the raw (unfiltered) signal
σ_{filtered}	Standard deviation of the Kalman-filtered signal
RMSE_{raw}	Root Mean Square Error of the raw sensor signal
$\text{RMSE}_{\text{filtered}}$	Root Mean Square Error of the filtered sensor signal
ϵ	Small positive constant to prevent division by zero

References

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